

orbital angular momentum = $\frac{nh}{2\pi}$

orbital $l(l+1) = \sqrt{l(l+1)} \frac{h}{2\pi}$

spin $s(s+1) = \sqrt{s(s+1)} \frac{h}{2\pi}$

$$\frac{h}{2\pi} = \hbar$$

Total spin = $\pm \frac{n}{2}$

$n \rightarrow$ unpaired e^-

$\uparrow \downarrow$

$\uparrow$

Total spin of Fe =

Fe = $4s^2 3d^6$

$n=4$

spin = $\pm \frac{4}{2} = \pm 2$

Spin multiplicity $\rightarrow (SM)$

No. of possible spin orientation present in atom

$SM = 2S + 1$ $S \rightarrow$ total spin (+ve)

e.g. $H \rightarrow 1s^1$ $\boxed{1}$ ✓

$SM = 2s + 1$
 $= 2(\frac{1}{2}) + 1$
 $= 1 + 1 = 2$

C $\rightarrow 1s^2 2s^2 2p^2$

$\boxed{\uparrow \uparrow}$

$SM = 2(\frac{1}{2} \times 2) + 1 = 2 + 1 = 3$

* Max^m spin multiplicity of subshell having $l=2$ d-orbital

$\boxed{\uparrow \uparrow \uparrow \uparrow \uparrow}$

$n=5$ $S = \frac{5}{2}$

$SM = 2(\frac{5}{2}) + 1 = 6$

Degeneracy →

Degeneracy

1 e⁻ system

→ Degeneracy define according to shell

Energy order →

1s < 2s ≈ 2p < 3s ≈ 3p ≈ 3d <

4s ≈ 4p ≈ 4d ≈ 4f

degeneracy

n=1 → 1

n=2 → 4

n=3 → 9

n=4 → 16

multi e⁻ system

→ Degeneracy define according to subshell

Energy order →

1s < 2s < 2p < 3s < 3p < 4s < 3d - - n² = 9

degeneracy

s → 1

p →  → 3

d →  → 5

f →  → 7

$\frac{E}{g}$

$E_n = \frac{E}{g}$

$\frac{-13.6 \times 1}{n^2} = \frac{E - 13.6}{g}$

n² = 9

n = 3

Schrodinger wave eqⁿ—

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2m}{h^2} (E - V) \psi = 0$$

$$\left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right) \psi + \frac{8\pi^2m}{h^2} (E - V) \psi = 0$$

$$\nabla^2 \psi + \frac{8\pi^2m}{h^2} (E - V) \psi = 0$$

$$(E - V) \psi = - \frac{\nabla^2 \psi h^2}{8\pi^2m}$$

$$E\psi - V\psi = - \frac{\nabla^2 h^2 \psi}{8\pi^2m}$$

$$E\psi = V\psi - \frac{\nabla^2 h^2}{8\pi^2m}$$

$$E\psi = \left(V - \frac{\nabla^2 h^2}{8\pi^2m} \right) \psi$$

$$\boxed{E\psi = \hat{H}\psi}$$

$$\hat{H} = V - \frac{\nabla^2 h^2}{8\pi^2m}$$

quantised

$\nabla \rightarrow$ Laplacian operator

$\hat{H} \rightarrow$ Hamiltonian operator

1) Continuous

2) Finite

3) Single value

4) $\int_{-\infty}^{+\infty} \psi^2 dx dy dz = 1$

ψ ✓

$\psi \rightarrow$ can have both +ve & -ve value.

$\psi^2 \rightarrow$ always +ve.

$\rightarrow$ Represent probability of finding e^- at a given point in space.

$\psi^2 \rightarrow \text{max}^m$ $\rightarrow$ Region is known as orbital

$\psi^2 = 0$ (min) $\rightarrow$ Region in which probability of finding e^- is zero $\rightarrow$ node

node $\rightarrow$ Radial node = $n - l - 1$ (RN)

$\rightarrow$ Angular node = l (AN)

Total node = $n - 1$

	R.N	A.N	
1s	0	0	
2s	1	0	
3s	2	0	
5s	4	0	1s 2s
2p	0	1	2p 3p
3p	1	1	3d 4d
4p	2	1	4f 5f
3d	0	2	
5d	2	2	
7f	3	3	

$$a) \psi_{2s} = \frac{1}{2\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(2 - \frac{r}{a_0}\right) e^{-r/a_0}$$

IF $r = r_0$ Radial node is formed
then

a) $r_0 = a_0$

b) $r_0 = a_0/2$

☒ c) $r_0 = 2a_0$

d) $r_0 = 4a_0$

$$\psi^2 = 0$$

$$\psi = 0$$

$$2 - \frac{r_0}{a_0} = 0$$

$$2 = \frac{r_0}{a_0}$$

$$r_0 = 2a_0$$

$$2) \psi = \frac{1}{16\sqrt{4}} \left(\frac{1}{a_0}\right)^{3/2} \left[(x-1) (x^2 - 8x + 12) \right] e^{-x/2}$$

$x = \frac{2r}{a_0}$ Calculate min & max^m
position of radial node
in terms of a_0

$$\psi = 0$$

$$x - 1 = 0$$

$$x = 1$$

$$\frac{2r}{a_0} = 1$$

$$r = a_0/2$$

$$x^2 - 8x + 12 = 0$$

$$x = 6 \quad x = 2$$

$$\frac{2r}{a_0} = 6$$

$$r = 3a_0$$

Shape of orbitals →

1) s-orbitals →

$$\rightarrow l=0$$

$$m=0$$

→ spherical shape



→ nodal plane = 0

2) p-orbital →

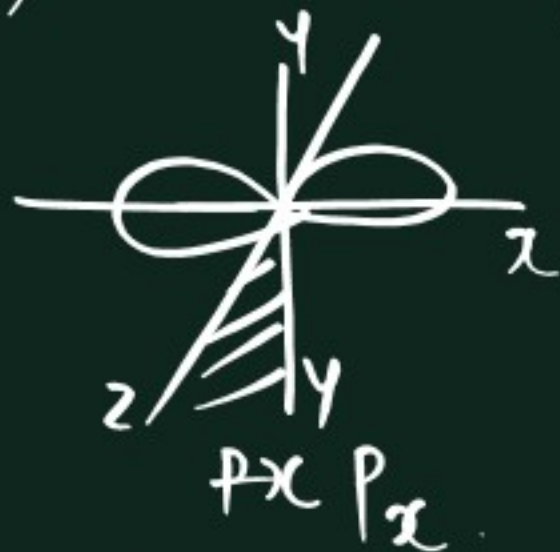
$$\rightarrow l=1$$

$$m = -1, 0, +1$$

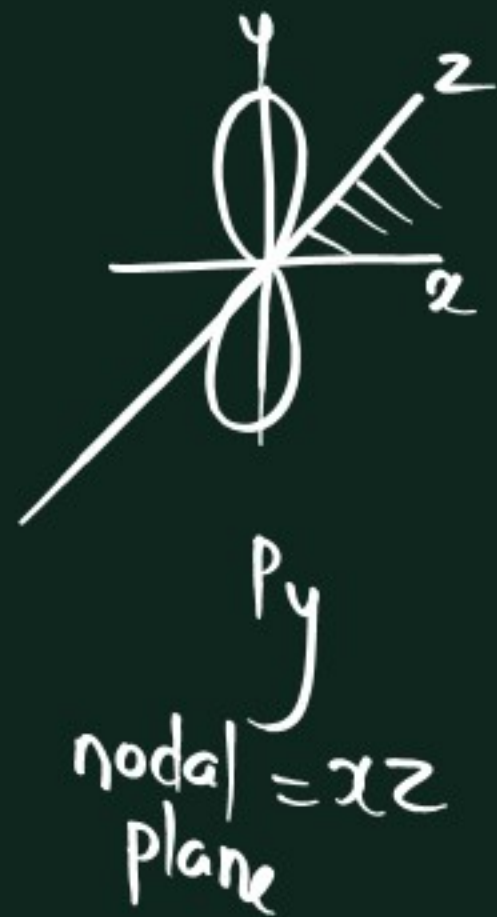
→

$$\begin{matrix} p_z & p_y & p_z \\ \downarrow & \downarrow & \downarrow \\ m=\pm 1 & m=\pm 1 & m=0 \end{matrix}$$

→ dumb-bell shape



nodal plane → yz



nodal plane = xz

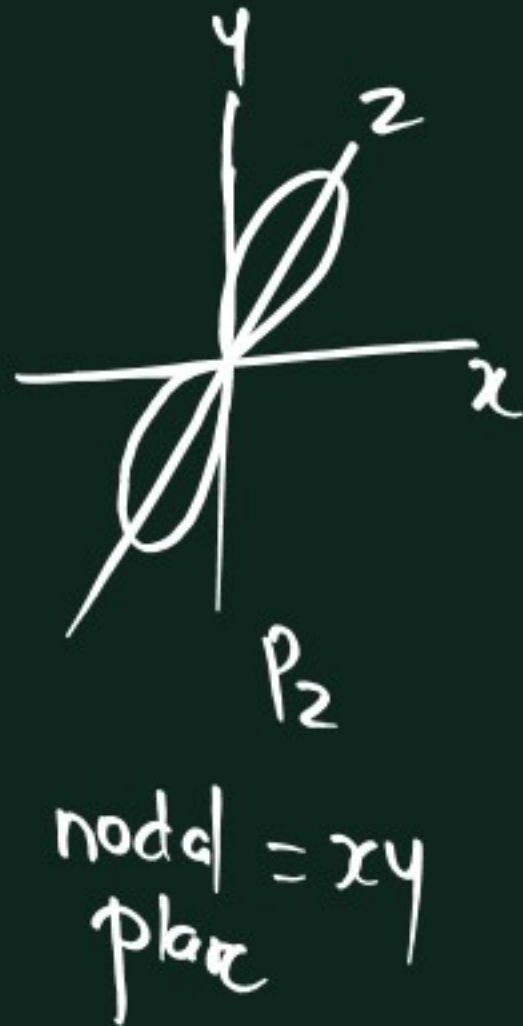
3) d-orbital →

$$l=2$$

$$m = -2, -1, 0, +1, +2$$

d →

$$\begin{matrix} d_{xy} & d_{yz} & d_{xz} & d_{x^2-y^2} & d_{z^2} \\ m=\pm 2 & m=\pm 1 & m=\pm 1 & m=\pm 2 & m=0 \end{matrix}$$



nodal plane = xy

→ double-dumb bell shape



dx_y

nodal plane → xy
 xz



dy_z

xy
 xz



dz_z

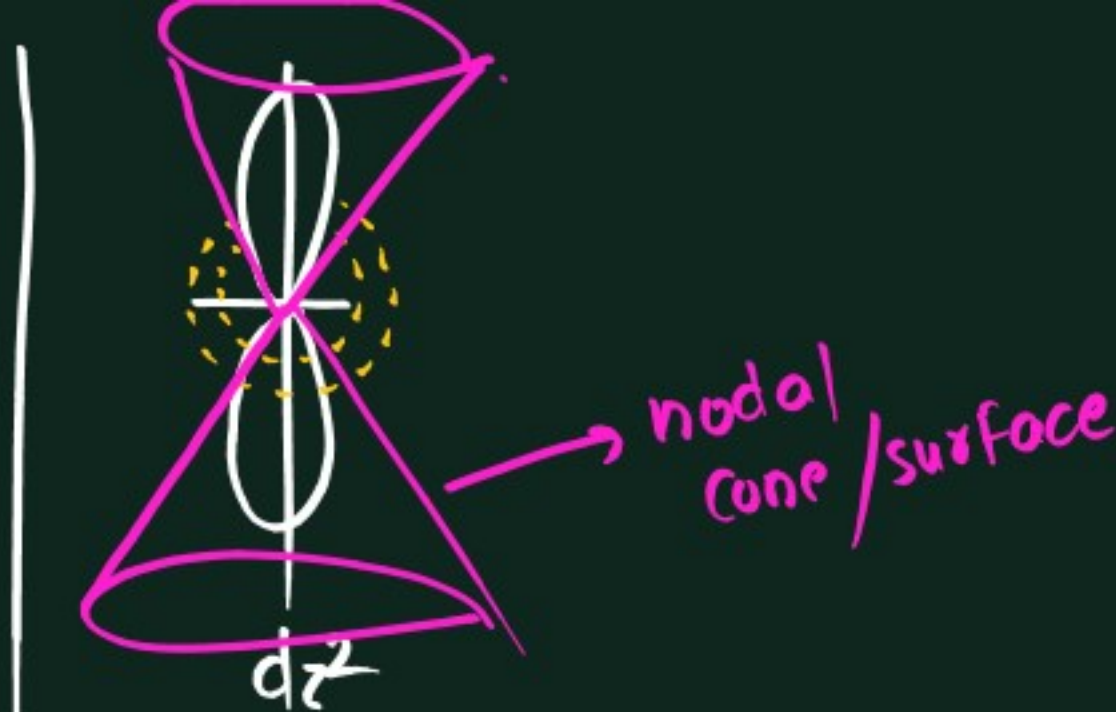
yz
 xy



dx^2-y^2

nodal

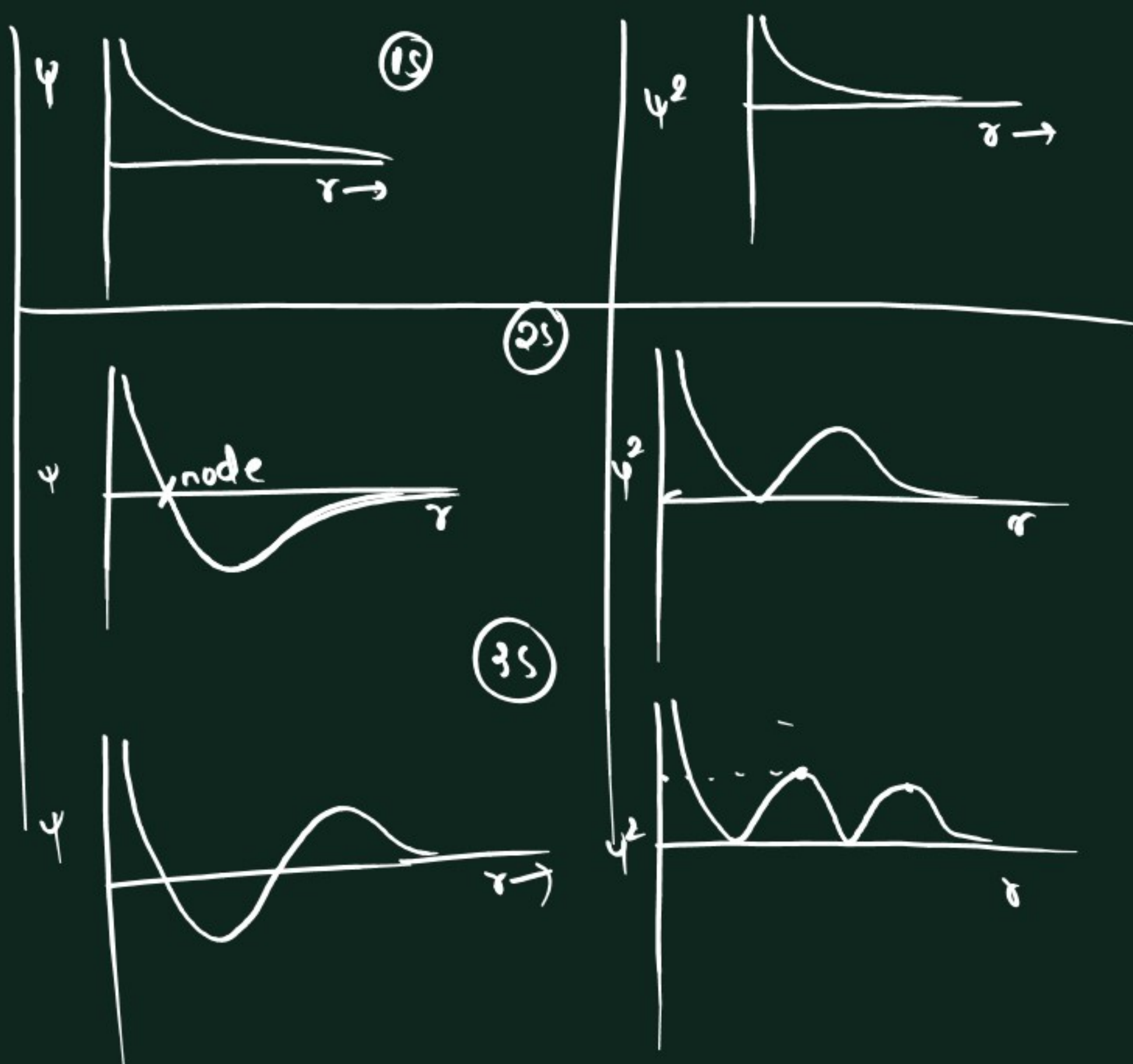
nodal plane → plane formed
line which
cuts x & y
axis at 45°



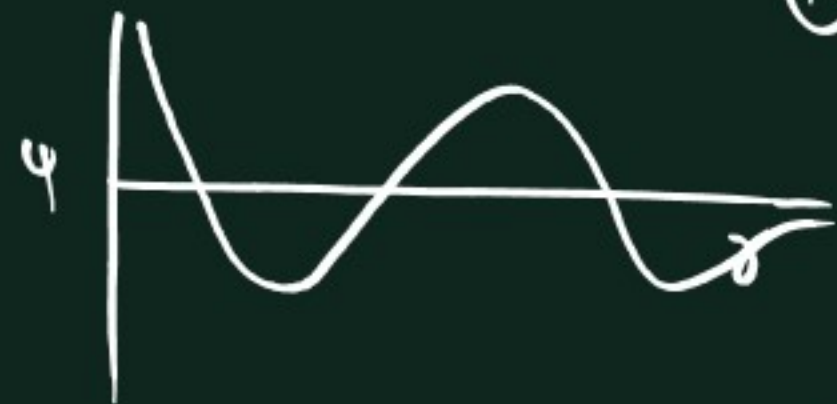
nodal
cone/surface

$\psi = R \times A$
 $\downarrow$ $\downarrow$
 Radial wave fn (n, l) angular wave fn (l, m)

ψ_r or $R \rightarrow$ Radial wave fn
 $\rightarrow$ variation of probability with radius



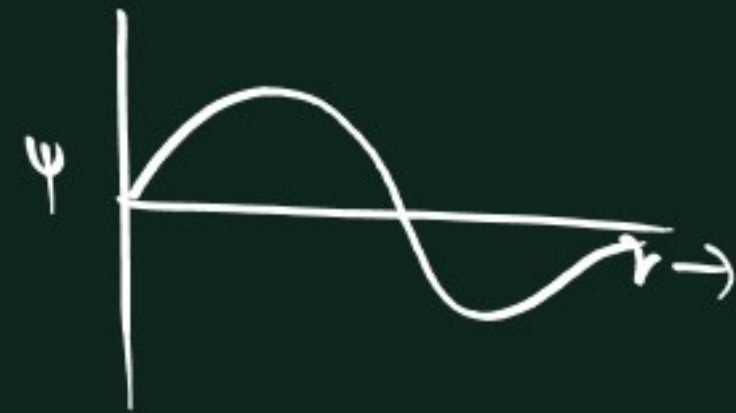
4s



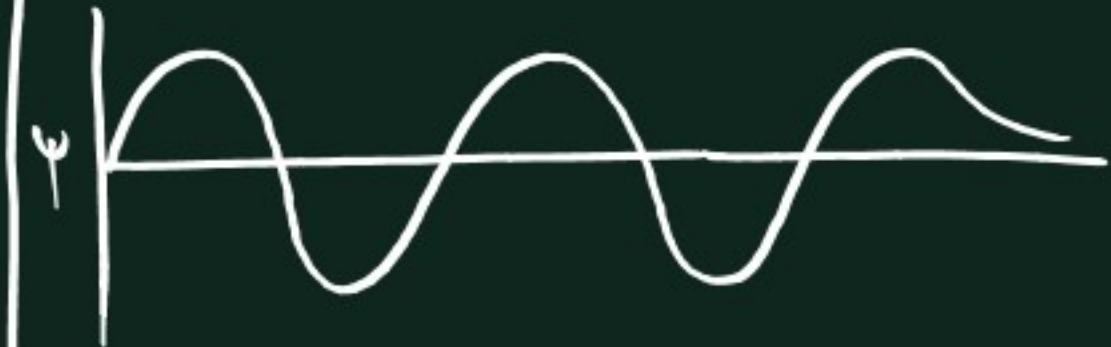
2p



3p



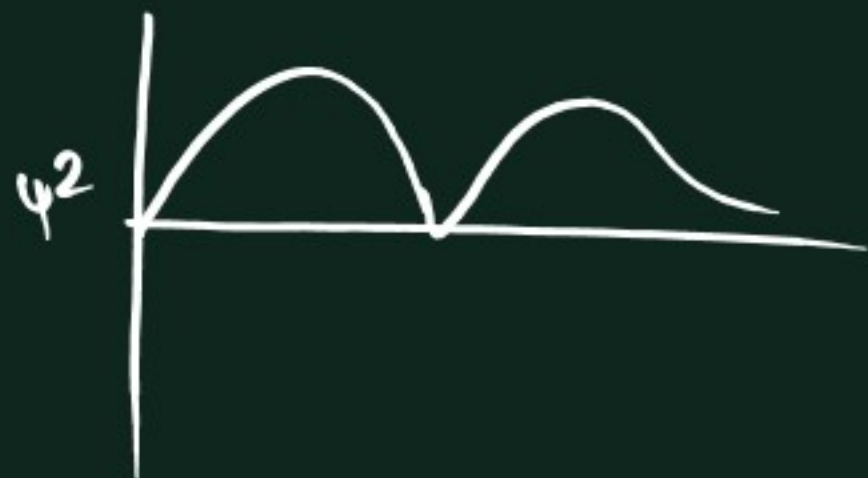
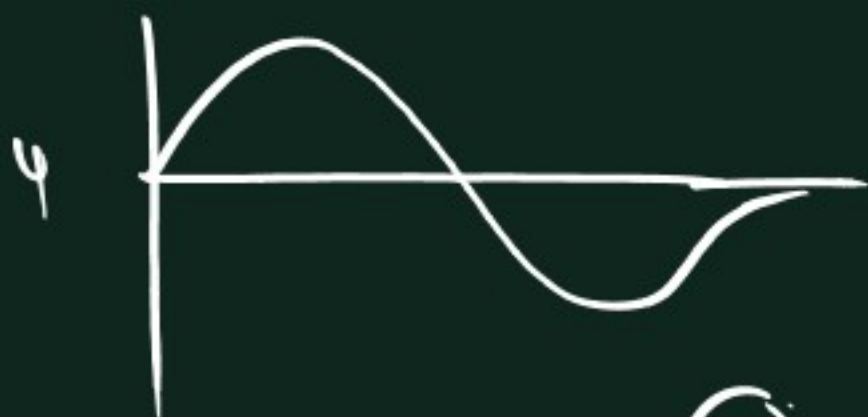
6p



3d



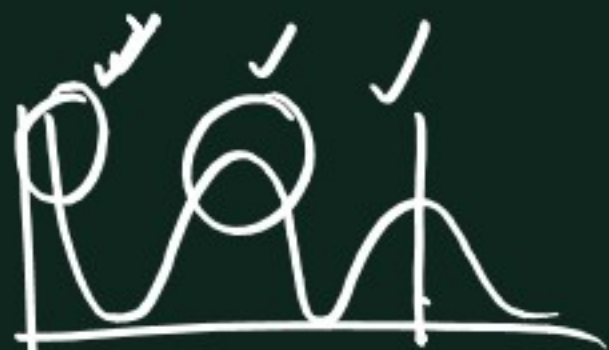
(4d)



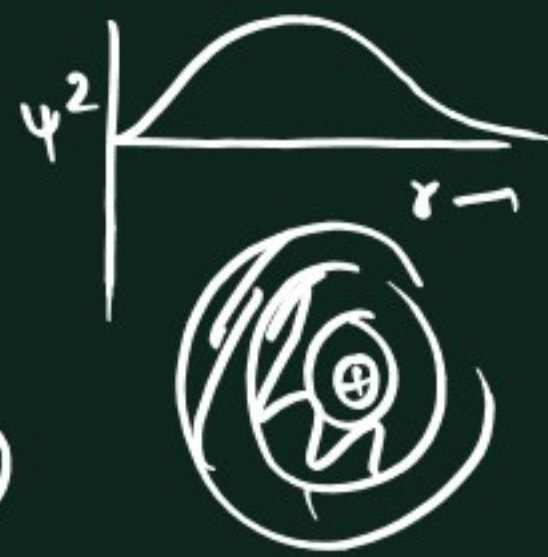
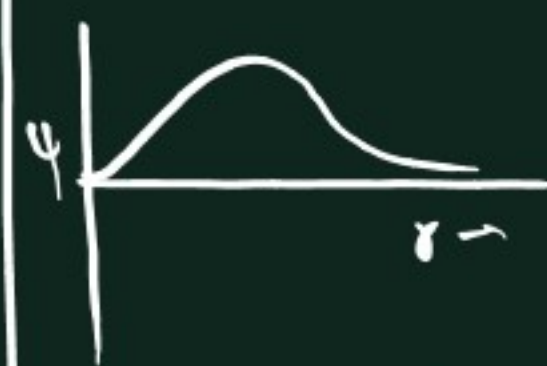
(4d)



(3)



(4f)



(5f)

$r_{max} \rightarrow$ dist'n at which probability is max^m

$\rightarrow r_{max} \rightarrow 1s < 2s < 3s$
 $2p < 3p < 4p$

$\rightarrow$ for some (n) $\rightarrow$

$r_{max} \rightarrow 3s > 3p > 3d$

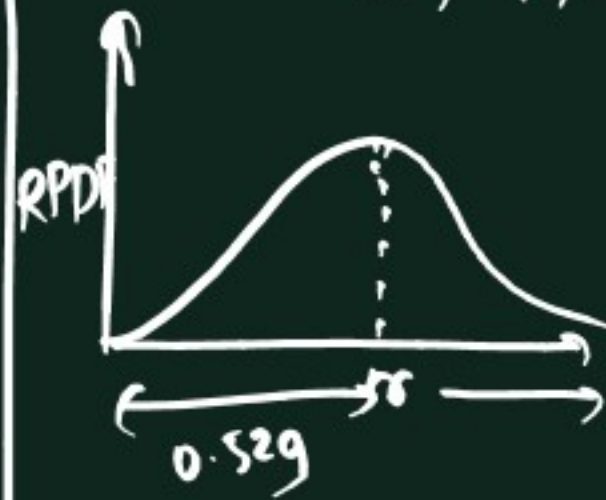
Penetration power $\rightarrow s > p > d > f$

RPDF → Radial probability distribution F^n
 → probability of finding e^- in small region of $4\pi r^2 dr$

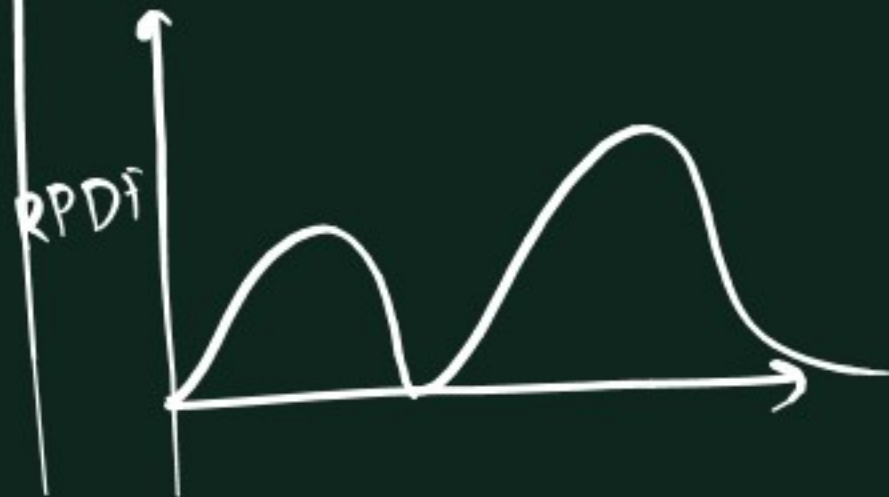
$$RPDF = \psi_r^2 dv = \psi_r^2 4\pi r^2 dr$$



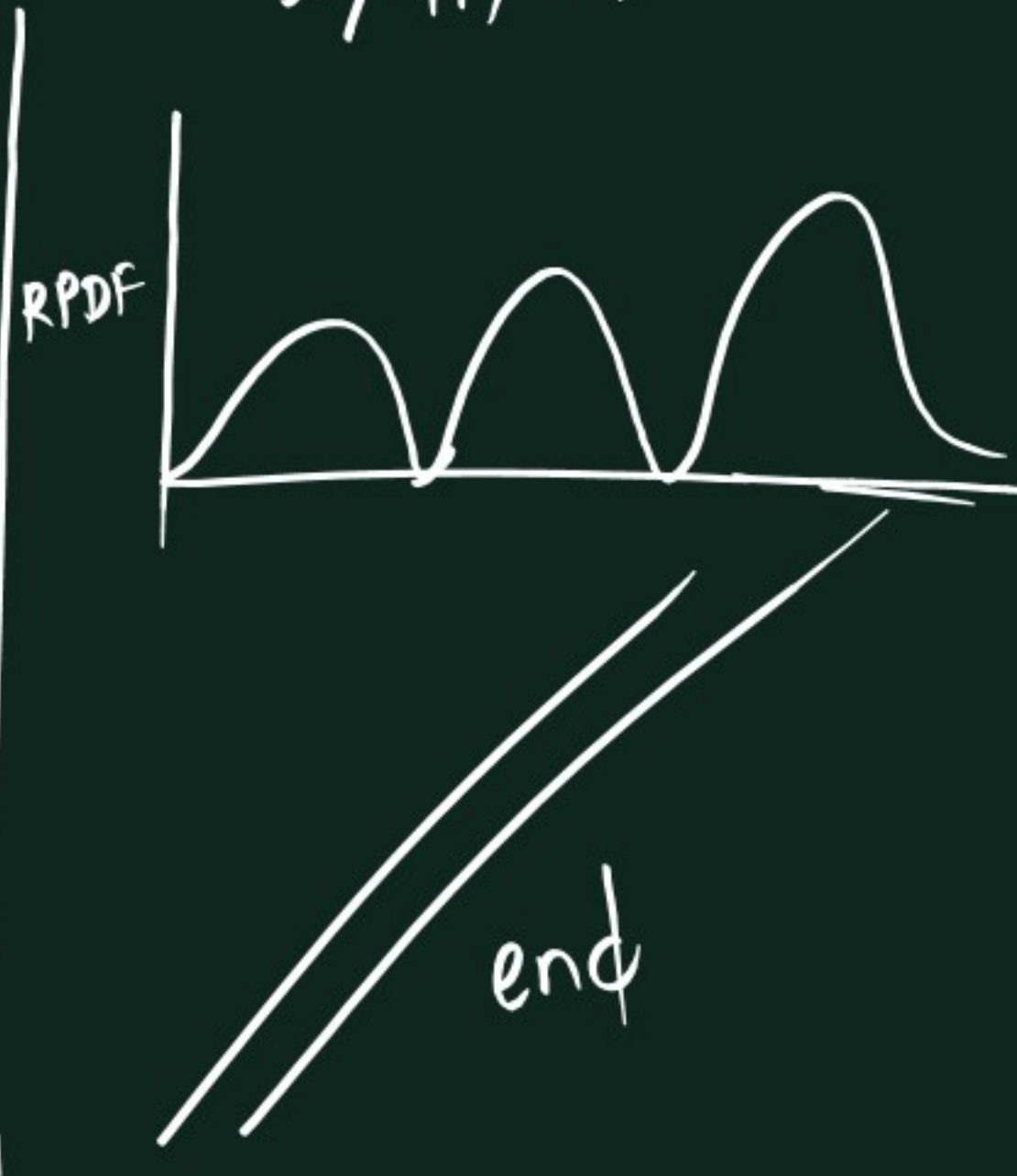
①s/2p/3d/4f



2s/3p/4d/5f



3s/4p/5d/6f



Inorganic Chemistry

Periodic Properties

* Dobereiner's triads

Triads $\rightarrow$ Li, Na, K

$$\text{Na} = \frac{\text{Li} + \text{K}}{2} = \frac{7 + 39}{2} = \frac{46}{2} = 23$$

2) Cl, Br, I

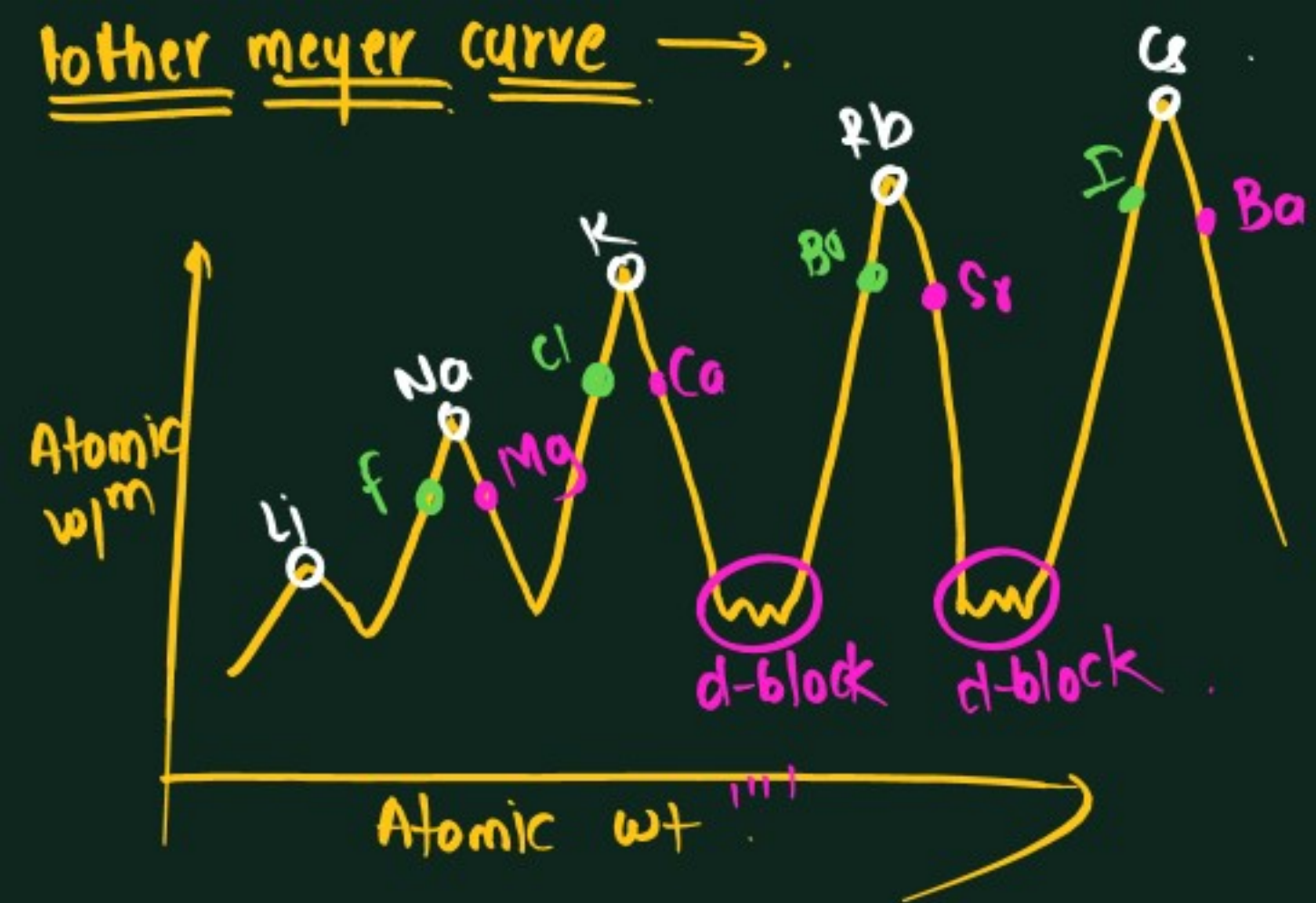
$$\text{Br} = \frac{\text{Cl} + \text{I}}{2} = \frac{35.5 + 127}{2} = \frac{162.5}{2} = 81.25$$

3) Ca, Sr, Ba.

Newland's octave rule $\rightarrow$

Li	Be	B	C	¹ N	² O	² F
⁸ Na	Mg	Al	Si	⁸ P	S	Cl
K	Ca					

lotther meyer curve →



$$d = \frac{\text{Atomic mass}}{\text{Atomic w/m}}$$